(DISCOMs). It has been steadily building captive generation capacity and, of late, buying power from energy exchanges.

Industry started investing in captive power generation in the 1970s, the time when State Electricity Boards (SEBs), the predecessors of DISCOMs, were diversifying towards agriculture and residential consumers, favouring them with lower tariffs. For the first time, industrial enterprises built about 2600 megawatts (MW) of captive power-generation capacity (1849 MW thermal and 559 MW diesel) during the Fifth Plan period in the 1970s. Captive power capacity had grown to around 17,000 MW by the end of the Ninth Plan in 2001–02. Investment in captive power plants received a big boost thereafter with capacity reaching as high as 58,000 MW at the end of financial year 2018–19.

Having installed large captive power generation capacity, the industry is buying a smaller share of power for manufacturing from DISCOMs. In 2017–18, captive power generated reached 180 billion units. Industry used a total of 469 billion that year, of which captive power comprised about 40 per cent. In fact, captive power generated by industry accounted for 16 per cent of the total power consumed in India in 2017–18. Excluding power captively produced and consumed in 2017–18, only 289 billion units of power were supplied by state utilities to industry, which was only about 25 per cent of total power supplied and used in India in 2017–18.

Electricity purchase from DISCOMs by industry is fast tapering off.

Two Major Transformations: From Fossil Fuels to Renewables, and Public to Private Sector

Total installed power-generation capacity in India (grid connected; not including power generated by gensets and renewable home lighting systems) at the end of October 2020 was 3.73 lakh MW or 373 GW. The generation capacity owned by the private sector was 1.75 lakh MW, over 47 per cent of the total power-generation capacity of the country.